Week 7 Decision Journal

Context

- The main priorities are achieving strong recall for ESI 2, 3 and 4, preventing dangerous undertriage, producing predictions quickly enough for real time use, and providing patient level explanations.
- This week's evaluation showed that improving one ESI category can worsen another. The ESI 2 focused HGB model achieved strong ESI 2 recall but reduced ESI 4 recall to 0.430. The priority was therefore revised to seek a safer balance across ESI 2, ESI 3 and ESI 4 rather than optimising one category in isolation.

Alternatives considered

- **Classweighted logistic regression with demographics and feature engineering:** This was the fastest and most transparent option because clinicians as these could help the model make a better decision, as it helps in real life triaging scenarios. It achieved ESI 2 recall of 0.785 and ESI 4 recall of 0.630, while keeping ESI 3 undertriage at 8.2%. However, it did not achieve the strongest combined ESI 2 and ESI 4 recall.
- **Stacking ensemble combining logistic regression, a neural network and HGB:** This achieved a macro F1 score of 0.484, ESI 2 recall of 0.786 and ESI 4 recall of 0.600. Its performance was relatively balanced, but it was the most complex option to explain, deploy, monitor and maintain.
- **Class weightings for ESI 2 on the HGB:** This achieved an ESI 2 recall of 0.797, but ESI 4 recall fell to 0.430.

Decision

Select the weighted Histogram Gradient Boosting model using {2: 2, 4: 1.5} class weights as the preferred weighting

Reasoning

- **It achieved the strongest combined ESI 2 recall and under triaging.** The model reduced ESI 2 undertriage to 19.6% whilst also maintaining relatively low ESI 3 and 4 undertriage levels
- **It meets the operational requirements.** The model produces predictions in approximately seconds and can generate a SHAP explanation in approximately one

second. As a single model, it is also easier to deploy and monitor than the stacking ensemble.

Things not yet known

- It is not yet known how to balance ESI 2, ESI 3 and ESI 4 simultaneously without improving one category by increasing undertriage or overtriage in another. Further testing should evaluate combinations of class weights and a cost matrix.
- It is not yet known whether XGBoost or CatBoost could provide a better balance across all three ESI categories.